

DATA SCIENCE

**With DATA ANALYTICS, MACHINE LEARNING,
NLP, DL & Gen AI using PYTHON & R**

INTRODUCTION TO DATA SCIENCE

- What is a Data Science?
- Who is a Data Scientist?
- Who can become a Data Scientist?
- What is an Artificial Intelligence?
- What is a Machine Learning?
- What is a Deep Learning?
- Artificial Intelligence Vs Machine Learning Vs Deep Learning
- Real Time Process of Data Science
- Data Science Real Time Applications
- Technologies used in Data Science
- Prerequisites Knowledge to Learn Data Science

INTRODUCTION TO MACHINE LEARNING

- What is a Machine Learning?
- Machine Learning Vs Statistics
- Traditional Programming Vs Machine Learning
- How Machine Will Learn like Human Learning
- Machine Learning Engineer Responsibilities
- Types of Machine Learning
 - Supervised learning
 - Un-Supervised learning
 - Reinforcement Learning

CORE PYTHON PROGRAMMING

- PYTHON Programming Introduction
- History of Python
- Python is Derived from?
- Python Features & Applications
- Why Python is Becoming Popular Now a Day?
- Existing Programming Vs Python Programming
- Writing Programs in Python
- Top Companies Using Python
- Python Programming Modes
 - Interactive Mode Programming
 - Scripting Mode Programming

- Flavors in Python, Python Versions
- Download & Install the Python in Windows & Linux
- How to set Python Environment in the System?
- Anaconda - Data Science Distributor
- Downloading and Installing Anaconda, Jupyter Notebook & Spyder
- Python IDE - Jupyter Notebook Environment
- Python IDE - Spyder Environment
- Python Identifiers(Literals), Reserved Keywords
- Variables, Comments, Lines and Indentations, Quotations
- Assigning Values to Variables
- Data Types in Python, Mutable Vs Immutable
- Fundamental Data Types: int, float, complex, bool, str
- Number Data Types: Decimal, Binary, Octal, Hexa Decimal & Number Conversions
- Inbuilt Functions in Python
- Data Type Conversions
- Priorities of Data Types in Python
- Python Operators
 - Arithmetic, Comparison (Relational), Assignment, Logical, Bitwise, Membership & Identity Operators
- Slicing & Indexing
 - Forward Direction Slicing with +ve Step
 - Backward Direction Slicing with -ve Step
- Decision Making Statements
 - If, if-else & elif Statements
- Looping Statements
 - Why we use Loops in python?
 - Advantages of Loops
 - for Loop & Nested for Loop
 - Using else Statement with for Loop
 - while Loop & Infinite while Loop
 - Using else with Python while Loop
- Conditional Statements
 - Break, Continue, Pass Statements

ADVANCED PYTHON PROGRAMMING

- Advanced Data Types: List, Tuple, Set, Frozenset, Dictionary, Range, Bytes & Bytearray, None
- List Data Structure
 - List indexing and splitting
 - Updating List values
 - List Operations
 - Iterating a List
 - Adding Elements to the List
 - Removing Elements from the List
 - List Built-in Functions & Methods

- Tuple Data Structure
 - Tuple Indexing and Splitting
 - Tuple Operations
 - Tuple Inbuilt Functions
 - Where use Tuple
 - List Vs Tuple
 - Nesting List and Tuple
- Set Data Structure
 - Creating a Set
 - Set Operations
 - Adding Items to the Set
 - Removing Items from the Set
 - Difference Between discard() and remove()
 - Union of Two Sets
 - Intersection of Two Sets
 - Difference of Two Sets
 - Set Comparisons
- Frozenset Data Structure
- Dictionary Data Structure
 - Creating & Accessing the Dictionary Values
 - Updating Dictionary Values
 - Deleting Elements Using del Keyword
 - Iterating Dictionary
 - Properties of Dictionary Keys
 - Built-in Dictionary Functions
 - Built-in Dictionary Methods
- List Vs Tuple Vs Set Vs Frozenset Vs Dict
- Range, Bytes, Bytearray & None
- Python Functions
 - Advantage of Functions in Python
 - Creating a Function
 - Function Calling
 - Parameters in Function
 - Call by Reference in Python
 - Types of Arguments
 - Required, Keyword, Default, Variable-Length Arguments
- Scope of Variables
- Python Built-in & Lambda Functions
- String with Functions
 - Strings Indexing and Splitting
 - String Operators
 - Python Formatting Operator
 - Built-in String Functions
- Python File Handling
 - Opening a File, Reading the File & Read Lines of the File
 - Looping through the File

- Writing the File & Creating a New File
- Using with Statement with Files
- File Pointer Positions
- Modifying File Pointer Position
- Renaming the File & Removing the File
- Writing Python Output to the Files
- File Related Methods
- Python Exceptions
 - Common Exceptions
 - Problem without Handling Exceptions
 - except Statement with no Exception
 - Declaring Multiple Exceptions
 - Finally Block
 - Raising & Custom Exceptions
- Python Packages
 - Python Libraries
 - Python Modules
 - Collection, Math, OS, Random, Statistics, Sys Modules
 - Date & Time Module
 - Loading the Module in our Python Code
 - import Statement
 - from-import Statement
 - Renaming a Module
- Regular Expressions
- Command Line Arguments
- Object Oriented Programming (OOPs)
 - Object-oriented vs Procedure-oriented Programming languages
 - Object, Class, Method, Inheritance, Polymorphism, Data Abstraction & Encapsulation
- Python Class and Objects
 - Creating Classes in Python
 - Creating an Instance of the Class
- Python Constructor
 - Creating the Constructor in Python
 - Parameterized & Non-Parameterized Constructors
 - In-built Class Functions & Attributes
- Python Inheritance
 - Python Multi-Level & Multiple Inheritance
 - Method Overriding
 - Data Abstraction in Python
- Graphical User Interface (GUI) Programming
- Python Tkinter
 - Tkinter Geometry
 - pack(), grid(), place() Methods
 - Tkinter Widgets

DATA ANALYSIS WITH PYTHON NUMPY

- NumPy Introduction
 - What is NumPy
 - The Need of NumPy
- NumPy Environment Setup
- N-Dimensional Array (Ndarray)
 - Creating a Ndarray Object
 - Finding the Dimensions of the Array
 - Finding the Size of Each Array Element
 - Finding the Data Type of Each Array Item
 - Finding the Shape and Size of the Array
 - Reshaping the Array Objects
 - Slicing in the Array
 - Finding the Maximum, Minimum, and Sum of the Array Elements
 - NumPy Array Axis
 - Finding Square Root and Standard Deviation
 - Arithmetic Operations on the Array
 - Array Concatenation
- NumPy Datatypes
 - NumPy dtype
 - Creating a Structured Data Type
- Numpy Array Creation
 - Numpy.empty
 - Numpy.Zeros
 - Numpy.ones
- Numpy Array from Existing Data
 - Numpy.asarray
- Numpy Arrays within the Numerical Range
 - Numpy.arrange
 - NumPy.linspace
 - Numpy.logspace
- NumPy Broadcasting
 - Broadcasting Rules
- NumPy Array Iteration
 - Order of Iteration
 - F-Style Order
 - C-Style Order
 - Array Values Modification
- NumPy String Functions
- NumPy Mathematical Functions
 - Trigonometric Functions
 - Rounding Functions
- NumPy Statistical functions
 - Finding the Min and Max Elements from the Array
 - Calculating Median, Mean, and Average of Array Items
- NumPy Sorting and Searching

- NumPy Copies and Views
- NumPy Matrix Library
- NumPy Linear Algebra
- NumPy Matrix Multiplication in Python

DATA ANALYSIS WITH PYTHON PANDAS

- Pandas Introduction & Pandas Environment Setup
 - Key Features of Pandas
 - Benefits of Pandas
 - Python Pandas Data Structure
 - Series
 - DataFrame
 - Panel
- Pandas Series
 - Creating a Series
 - Create an Empty Series
 - Create a Series using Inputs
 - Accessing Data from Series with Position
 - Series Object Attributes
 - Retrieving Index Array and Data Array of a Series Object
 - Retrieving Types (dtype) and Size of Type (itemsize)
 - Retrieving Shape
 - Retrieving Dimension, Size and Number of Bytes
 - Checking Emptiness and Presence of NaNs
 - Series Functions
- Pandas DataFrame
 - Create a DataFrame
 - Create an Empty DataFrame
 - Create a DataFrame using Inputs
- Column Selection, Addition & Deletion
- Row Selection, Addition & Deletion
- DataFrame Functions
- Merging, Joining & Combining DataFrames
- Pandas Concatenation
- Pandas Time Series
 - Datetime
 - Time Offset
 - Time Periods
 - Convert String to Date
- Viewing/Inspecting Data (loc & iloc)
- Data Cleaning
- Filter, Sort, and Groupby
- Statistics on DataFrame
- Pandas Vs NumPy
- DataFrame Plotting
 - Line: Line Plot (Default)

- Bar: Vertical Bar Plot
- Barh: Horizontal Bar Plot
- Hist: Histogram Plot
- Box: Box Plot
- Pie: Pie Chart
- Scatter: Scatter Plot

DBMS - Structured Query Language

- Introduction & Models of DBMS
- SQL & Sub Language of SQL
- Data Definition Language (DDL)
- Data Manipulation Language (DML)
- Data Query/Retrieval Language (DQL/DRL)
- Transaction Control Language (TCL)
- Data Control Language (DCL)
- Installation of MySQL & Database Normalization
- Sub Queries & Key Constraints
- Aggregative Functions, Clauses & Views

Importing & Exporting Data

- Data Extraction from CSV (pd.read_csv)
- Data Extraction from TEXT File (pd.read_table)
- Data Extraction from CLIPBOARD (pd.read_clipboard)
- Data Extraction from EXCEL (pd.read_excel)
- Data Extraction from URL (pd.read_html)
- Writing into CSV (df.to_csv)
- Writing into EXCEL (df.to_excel)
- Data Extraction from DATABASES
 - Python MySQL Database Connection
 - Import mysql.connector Module
 - Create the Connection Object
 - Create the Cursor Object
 - Execute the Query

DATA VISUALIZATION WITH PYTHON MATPLOTLIB

- Data Visualization Introduction
- Tasks of Data Visualization
- Benefit of Data Visualization
- Plots for Data Visualization
- Matplotlib Architecture
- General Concept of Matplotlib
- Matplotlib Environment Setup
- Verify the Matplotlib Installation
- Working with PyPlot
- Formatting the Style of the Plot
- Plotting with Categorical Variables

- Multi-Plots with Subplot Function
- Line Graph
- Bar Graph
- Histogram
- Scatter Plot
- Pie Plot
- 3Dimensional - 3D Graph Plot
- mpl_toolkits
- Functions of Matplotlib
- Contour Plot, Quiver Plot, Violin Plot
- 3D Contour Plot
- 3D Wireframe Plot
- 3D Surface Plot
- Box Plot
 - What is a Boxplot?
 - Mean, Median, Quartiles, Outliers
 - Inter Quartile Range (IQR), Whiskers
 - Data Distribution Analysis
 - Boxplot on a Normal Distribution
 - Probability Density Function
 - 68–95–99.7 Rule (Empirical rule)

Data Analysis Project using Python Programming

MACHINE LEARNING

- What is Machine Learning
- Importance of Machine Learning
- Need for Machine Learning
- Statistics Vs Machine Learning
- Traditional Programming Vs Machine Learning
- How Machine Learning like Human Learning
- How does Machine Learning Work?
- Machine Learning Engineer Responsibilities
- Life Cycle of Machine Learning
 - Gathering Data
 - Data preparation
 - Data Wrangling
 - Analyze Data
 - Train the model
 - Test the model
 - Deployment
- Features of Machine Learning

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- History of Machine Learning
- Applications of Machine Learning
- Types of Machine Learning
 - Supervised Machine Learning
 - Unsupervised Machine Learning
 - Reinforcement Learning

Supervised Machine Learning

- How Supervised Learning Works?
- Steps Involved in Supervised Learning
- Types of supervised Machine Learning Algorithms
 - Classification
 - Regression
- Advantages of Supervised Learning
- Disadvantages of Supervised Learning

Unsupervised Machine Learning

- How Unsupervised Learning Works?
- Why use Unsupervised Learning?
- Types of Unsupervised Learning Algorithm
 - Clustering
 - Association
- Advantages of Unsupervised Learning
- Disadvantages of Unsupervised Learning
- Supervised Vs Unsupervised Learning
- Reinforcement Machine Learning
- How to get Datasets for Machine Learning?
 - What is a Dataset?
 - Types of Data in Datasets
 - Popular Sources for Machine Learning Datasets

Data Preprocessing in Machine Learning

- Why do we need Data Preprocessing?
 - Getting the Dataset
 - Importing Libraries
 - Importing Datasets
 - Finding Missing Data
 - By Deleting the Particular Row
 - By Calculating the Mean
 - Encoding Categorical Data
 - LabelEncoder
 - OneHotEncoder
 - Splitting Dataset into Training and Test Set
 - Feature Scaling
 - Standardization
 - Normalization

Classification Algorithms in Machine Learning

- What is the Classification Algorithm?
- Types of Classifications
 - Binary Classifier
 - Multi-class Classifier
- Learners in Classification Problems
 - Lazy Learners
 - Eager Learners
- Types of ML Classification Algorithms
 - Linear Models
 - Logistic Regression
 - Support Vector Machines
 - Non-linear Models
 - K-Nearest Neighbors
 - Naïve Bayes
 - Decision Tree Classification
 - Random Forest Classification
 - Kernel SVM
- Evaluating a Classification Model
 - Confusion Matrix
 - What is a Confusion Matrix?
 - True Positive
 - True Negative
 - False Positive – Type 1 Error
 - False Negative – Type 2 Error
 - Why need a Confusion matrix?
 - Precision
 - Recall
 - Precision vs Recall
 - F1-score
 - Confusion Matrix in Scikit-Learn
 - Confusion Matrix for Multi-Class Classification
 - Log Loss or Cross-Entropy Loss
 - AUC-ROC curve
- Use cases of Classification Algorithms

K-Nearest Neighbor(KNN) Algorithm in Machine Learning

- Why do we Need a K-NN Algorithm?
- How does K-NN work?
 - What is Euclidean Distance
 - How it Calculates the Distance
- How to Select the Value of K in the K-NN Algorithm?
- Advantages of KNN Algorithm
- Disadvantages of KNN Algorithm
- Python Implementation of the KNN Algorithm
- Analysis on Social Network Ads Dataset

- Steps to Implement the K-NN Algorithm
 - Data Pre-processing Step
 - Fitting the K-NN algorithm to the Training Set
 - Predicting the Test Result
 - Test Accuracy of the Result (Creation of Confusion Matrix)
 - Visualizing the Test Set Result.
 - Improve the Performance of the K-NN Model

Naïve Bayes Classifier Algorithm in Machine Learning

- Why is it Called Naïve Bayes?
 - Naïve Means?
 - Bayes Means?
- Bayes' Theorem
 - Posterior Probability
 - Likelihood Probability
 - Prior Probability
 - Marginal Probability
- Working of Naïve Bayes' Classifier
- Advantages of Naïve Bayes Classifier
- Disadvantages of Naïve Bayes Classifier
- Applications of Naïve Bayes Classifier
- Types of Naïve Bayes Model
 - Gaussian Naïve Bayes Classifier
 - Multinomial Naïve Bayes Classifier
 - Bernoulli Naïve Bayes Classifier
- Python Implementation of the Naïve Bayes Algorithm
- Steps to Implement the Naïve Bayes Algorithm
 - Data Pre-processing Step
 - Fitting Naive Bayes to the Training set
 - Predicting the Test Result
 - Test Accuracy of the Result (Creation of Confusion matrix)
 - Visualizing the Test Set Result
 - Improve the Performance of the Naïve Bayes Model

Decision Tree Classification Algorithm in Machine Learning

- Why use Decision Trees?
- Types of Decision Trees
 - Categorical Variable Decision Tree
 - Continuous Variable Decision Tree
- Decision Tree Terminologies
- How does the Decision Tree Algorithm Work?
- Attribute Selection Measures
 - Entropy
 - Information Gain
 - Gini index
 - Gain Ratio

- Algorithms used in Decision Trees
 - ID3 Algorithm → (Extension of D3)
 - C4.5 Algorithm → (Successor of ID3)
 - CART Algorithm → (Classification & Regression Tree)
- How to Avoid/Counter Overfitting in Decision Trees?
 - Pruning Decision Trees
 - Random Forest
- Pruning: Getting an Optimal Decision tree
- Advantages of the Decision Tree
- Disadvantages of the Decision Tree
- Python Implementation of Decision Tree
- Steps to Implement the Decision Tree Algorithm
 - Data Pre-processing Step
 - Fitting a Decision-Tree Algorithm to the Training Set
 - Predicting the Test Result
 - Test Accuracy of the Result (Creation of Confusion matrix)
 - Visualizing the Test Set Result
 - Improve the Performance of the Decision Tree Model

Random Forest Classifier Algorithm in Machine Learning

- Working of the Random Forest Algorithm
- Assumptions for Random Forest
- Why use Random Forest?
- How does Random Forest Algorithm Work?
 - Ensemble Techniques
 - Bagging (Bootstrap Aggregation)
- Applications of Random Forest
- Disadvantages of Random Forest
- Python Implementation of Random Forest Algorithm
- Steps to Implement the Random Forest Algorithm:
 - Data Pre-processing Step
 - Fitting the Random Forest Algorithm to the Training Set
 - Predicting the Test Result
 - Test Accuracy of the Result (Creation of Confusion Matrix)
 - Visualizing the Test Set Result
 - Improving the Performance of the Random Forest Model

Logistic Regression Algorithm in Machine Learning

- Logistic Function (Sigmoid Function)
- Assumptions for Logistic Regression
- Logistic Regression Equation
- Type of Logistic Regression
 - Binomial Logistic Regression
 - Multinomial Logistic Regression
 - Ordinal Logistic Regression
- Python Implementation of Logistic Regression (Binomial)

- Steps to Implement the Logistic Regression:
 - Data Pre-processing Step
 - Fitting Logistic Regression to the Training Set
 - Predicting the Test Result
 - Test Accuracy of the Result (Creation of Confusion Matrix)
 - Visualizing the Test Set Result
 - Improve the Performance of the Logistic Regression Model

Support Vector Machine Algorithm

- Types of Support Vector Machines
 - Linear Support Vector Machine
 - Non-Linear Support Vector Machine
- Hyperplane in the SVM Algorithm
- Support Vectors in the SVM Algorithm
- How does SVM Works?
 - How does Linear SVM Works?
 - How does Non-Linear SVM Works?
- Python Implementation of Support Vector Machine
- Steps to Implement the Support Vector Machine:
 - Data Pre-processing Step
 - Fitting Support Vector Machine to the Training Set
 - Predicting the Test Result
 - Test Accuracy of the Result (Creation of Confusion Matrix)
 - Visualizing the Test Set Result
 - Improve the Performance of the Support Vector Machine Model

Regression Algorithms in Machine Learning

- Terminologies Related to the Regression Analysis
 - Dependent Variable
 - Independent Variable
 - Outliers
 - Multi-collinearity
 - Under fitting and Overfitting
- Why do we use Regression Analysis?
- Types of Regression
 - Linear Regression
 - Logistic Regression
 - Polynomial Regression
 - Support Vector Regression
 - Decision Tree Regression
 - Random Forest Regression
 - Ridge Regression
 - Lasso Regression

Linear Regression in Machine Learning

- Types of Linear Regression
 - Simple Linear Regression
 - Multiple Linear Regression
- Linear Regression Line
 - Positive Linear Relationship
 - Negative Linear Relationship
- Finding the Best Fit Line
 - Cost Function
 - Gradient Descent
 - Model Performance
 - R-Squared Method
- Assumptions of Linear Regression

Simple Linear Regression in Machine Learning

- SLR Model
- Implementation of Simple Linear Regression Algorithm using Python
 - Data Pre-processing Step
 - Fitting Simple Linear Regression to the Training Set
 - Predicting the Test Result
 - Test Accuracy of the
 - Visualizing the Test Set Result.
 - Try to Improve the Performance of the Model

Multiple Linear Regression in Machine Learning

- MLR Equation
- Assumptions for Multiple Linear Regression
- Implementation of Multiple Linear Regression model using Python
 - Data Pre-processing Step
 - Fitting Multiple Linear Regression to the Training Set
 - Predicting the Test Result
 - Test Accuracy of the
 - Visualizing the Test Set Result.
 - Try to Improve the Performance of the Model

Backward Elimination

- What is Backward Elimination?
- Steps of Backward Elimination
- Need for Backward Elimination: An optimal Multiple Linear Regression model
- Implement the Steps for Backward Elimination method

Polynomial Regression in Machine Learning

- Need for Polynomial Regression
- Equation of the Polynomial Regression Model
- Implementation of Polynomial Regression using Python

- Steps for Polynomial Regression:
 - Data Pre-processing
 - Build a Linear Regression Model
 - Build a Polynomial Regression Model
 - Visualize the Result for Linear Regression Model
 - Visualize the Result for Polynomial Regression Model
 - Predicting the Final Result with the Linear Regression Model
 - Predicting the Final Result with the Polynomial Regression Model
- Support Vector Regression (SVR)
- Decision Tree Regression
- Random Forest Regression
- Ridge Regression
- Lasso Regression
- Linear Regression Vs Logistic Regression
- Classification vs Regression

Clustering Algorithms in Machine Learning

- Types of Clustering Methods
 - Partitioning Clustering
 - Density-Based Clustering
 - Distribution Model-Based Clustering
 - Hierarchical Clustering
 - Fuzzy Clustering
- Clustering Algorithms
 - K-Means Algorithm
 - Mean-shift Algorithm
 - DBSCAN Algorithm
 - Expectation-Maximization Clustering using GMM
 - Agglomerative Hierarchical Algorithm
 - Affinity Propagation
- Applications of Clustering

Hierarchical Clustering Algorithm in Machine Learning

- Hierarchical Clustering Technique Approaches
- Why Hierarchical Clustering?
- Agglomerative Hierarchical Clustering
- How the Agglomerative Hierarchical Clustering Work?
- Measure for the Distance between two Clusters
 - Single Linkage
 - Complete Linkage
 - Average Linkage
 - Centroid Linkage
- Working of Dendrogram in Hierarchical Clustering
- Hierarchical Clustering Example with Scratch Data
- Python Implementation of Agglomerative Hierarchical Clustering

- Steps for Implementation of Agglomerative Hierarchical Clustering using Python
 - Data Pre-processing
 - Finding the Optimal Number of Clusters using the Dendrogram
 - Training the Hierarchical Clustering Model
 - Visualizing the Clusters

K-Means Clustering Algorithm in Machine Learning

- What is K-Means Algorithm?
- How does the K-Means Algorithm Work?
- How to Choose the Value of "K Number of Clusters" in K-Means Clustering?
 - Elbow Method
 - Within Cluster Sum of Squares (WCSS)
- K-Means Clustering Example with Scratch Data
- Python Implementation of K-means Clustering Algorithm
- Steps to Implement of K-means Clustering Algorithm
 - Data Pre-processing
 - Finding the Optimal Number of Clusters using the Elbow Method
 - Training the K-means Algorithm on the Training Dataset
 - Visualizing the Clusters

Association Rules in Machine Learning

- Association Rules
- Pattern Detection
- Market Basket Analysis
- Support, Confidence, Expected Confidence, Lift
- Finding Item Sets with High Support
- Finding Item Rules with High Confidence or Lift

Apriori Algorithm in Machine Learning

- Apriori Algorithm
- How does Apriori Algorithm Works?
- Apriori Algorithm Example
- Implementation of Apriori Algorithm using Python
- Limitations of Apriori Algorithm

Dimensionality Reduction & Model Selection Boosting

- Dimensionality Reduction
 - Principal Component Analysis (PCA)
 - Linear Discriminant Analysis (LDA)
 - Kernel PCA
- Model Selection Boosting
 - Model Selection
 - Grid Search & K-Fold Cross Validation
 - XGBoost

STATISTICS

- Mean, Median and Mode
- Data Variability, Range, Quartiles
- IQR, Calculating Percentiles
- Variance, Standard Deviation, Statistical Summaries
- Types of Distributions – Normal, Binomial, Poisson
- Probability Distributions & Skewness
- Data Distribution, 68–95–99.7 rule (Empirical rule)
- Descriptive Statistics and Inferential Statistics
- Statistics Terms and Definitions, Types of Data
- Data Measurement Scales, Normalization, Standardization
- Measure of Distance, Euclidean Distance
- Probability Calculation – Independent & Dependent
- Entropy, Information Gain
- Regression

NATURAL LANGUAGE PROCESSING

- Natural Language Processing Introduction
 - What is NLP?
 - History of NLP
 - Advantages of NLP
 - Disadvantages of NLP
- Components of NLP
 - Natural Language Understanding (NLU)
 - Natural Language Generation (NLG)
 - Difference between NLU and NLG
- Applications of NLP
- How to build an NLP Pipeline?
- Phases of NLP
 - Lexical Analysis and Morphological
 - Syntactic Analysis (Parsing)
 - Semantic Analysis
 - Discourse Integration
 - Pragmatic Analysis
- Why NLP is Difficult?
- NLP APIs
- NLP Libraries
- Natural Language Vs Computer Language

Exploring Features of NLTK

- Open the Text File for Processing
- Import Required Libraries
- Sentence Tokenizing
- Word Tokenizing
- Find the Frequency Distribution
- Plot the Frequency Graph

- Remove Punctuation Marks
- Plotting Graph without Punctuation Marks
- List of Stopwords
- Removing Stopwords
- Final Frequency Distribution
- Word Cloud
 - Word Cloud Properties
 - Python Code Implementation of the Word Cloud
 - Word Cloud with the Circle Shape
 - Word Cloud Advantages
 - Word Cloud Disadvantages
- Stemming
 - Stemmer Examples
 - Stemming Algorithms
 - Porter's Stemmer
 - Lovin's Stemmer
 - Dawson's Stemmer
 - Krovetz Stemmer
 - Xerox Stemmer
 - Snowball Stemmer
- Lemmatization
 - Difference between Stemmer and Lemmatizer
 - Demonstrating how a lemmatizer works
 - Lemmatizer with default PoS value
 - Demonstrating the power of lemmatizer
 - Lemmatizer with different POS values
- Part-of-Speech (PoS) Tagging
 - Why do we need Part of Speech (POS)?
 - Part of Speech (PoS) Tags
- Chunking
 - Categories of Phrases
 - Phrase Structure Rules
- Chinking
- Named Entity Recognition (NER)
 - Use-Cases
 - Commonly used Types of Named Entity
- WordNet
- Bag of Words
 - What is the Bag-of-Words method?
 - Creating a basic Structure on Sentences
 - Words with Frequencies
 - Combining all the Words
 - Final Model of our Bag of Words
 - Applications & Limitations
- TF-IDF
 - Term Frequency

- Inverse Document Frequency
- Term Frequency - Inverse Document Frequency

Deploying a Machine Learning Model on a Web using Flask

- What is Model Deployment?
- What is Flask?
- Installing Flask on your Machine
- Understanding the Problem Statement
- Build our Machine Learning Model
- Create the Webpage
- Connect the Webpage with the Model
- Working of the Deployed Model

DEEP LEARNING INTRODUCTION

- What is Deep Learning?
- Deep learning Process
- Types of Deep Learning Networks
 - Deep Neural Networks
 - Artificial Neural Networks
 - Convolutional Neural Networks
 - Recurrent Neural Networks
- TensorFlow
 - History of TensorFlow
 - Components of TensorFlow
 - Use Cases/Applications of TensorFlow
 - Features of TensorFlow
- Installation of TensorFlow through pip & conda
- Advantage and Disadvantage of TensorFlow
- TensorFlow Playground
- Introduction to Keras, OpenCV & Theano
- Implementation of Deep Learning

ARTIFICIAL INTELLIGENCE INTRODUCTION

- What is Artificial Intelligence?
 - Why Artificial Intelligence?
 - Goals of Artificial Intelligence
 - What Comprises to Artificial Intelligence?
 - Advantages of Artificial Intelligence
 - Disadvantages of Artificial Intelligence
- Applications of Artificial Intelligence
- History of Artificial Intelligence
- Types of Artificial Intelligence
- Types of AI Agents
 - Simple Reflex Agent

- Model-Based Reflex Agent
- Goal-Based Agents
- Utility-Based Agent
- Learning Agent
- Search Algorithms in Artificial Intelligence
 - Search Algorithm Terminologies
 - Properties of Search Algorithms
 - Types of Search Algorithms
- Subsets of Artificial Intelligence
- Implementation of Artificial Intelligence

R PROGRAMMING

- Why R Programming is Important?
- Why Learn R?
- History of Python
- Features of R
- Applications of R
- Comparison between R and Python
- Which is Better to Choose
- Pros and Cons of R
- Companies using R
- R Packages
- Downloading and Installing R
- What is CRAN?
- Setting R Environment:
 - Search Packages in R Environment
 - Search Packages in Machine with inbuilt function and manual searching
 - Attach Packages to R Environment
 - Install Add-on Packages from CRAN
 - Detach Packages from R Environment
 - Functions and Packages Help
- R Programming IDE
 - RStudio
 - Downloading and Installing RStudio
- Variable Assignment
 - Displaying Variables
 - Deleting Variables
- Comments
 - Single Line
 - Multi Line Comments
- Data Types
 - Logical
 - Integer
 - Double

- Complex
 - Character
- Operators
 - Arithmetic, Relational, Logical, Assignment Operators
 - R as Calculator
 - Performing different Calculations
- Functions
 - Inbuilt Functions
 - User Defined Functions
- STRUCTURES
 - Vector
 - List
 - Matrix
 - Data frame
 - Array
 - Factors
- Inbuilt Constants & Functions
- Vectors
 - Vector Creation
 - Single Element Vector
 - Multiple Element Vector
 - Vector Manipulation
 - Sub setting & Accessing the Data in Vector
- Lists
 - Creating a List
 - Naming List Elements
 - Accessing List Elements
 - Manipulating List Elements
 - Merging Lists
 - Converting List to Vector
- Matrix
 - Creating a Matrix
 - Accessing Elements of a Matrix
 - Matrix Manipulations
 - Dimensions & Transpose of Matrix
- Data Frames
 - Create Data Frame
 - Vector to Data Frame
 - Character Data Converting into Factors: StringsAsFactors
 - Convert the columns of a data frame to characters
 - Extract Data from Data Frame
 - Expand Data Frame, Column Bind and Row Bind
- Merging / Joining Data Frames
 - Inner, Outer & Cross Join
- Arrays
 - Create Array with Multiple Dimensions

- Naming Columns and Rows
- Accessing Array Elements
- Manipulating Array Elements
- Calculations across Array Elements
- Factors
 - Factors in Data Frame
 - Changing the Order of Levels
 - Generating & Deleting Factor Levels

Loading and Reading Data in R

- Data Extraction from CSV
 - Getting and Setting the Working Directory
 - Input as CSV File, Reading a CSV File
 - Analyzing the CSV File, Writing into a CSV File
- Data Extraction from URL
- Data Extraction from CLIPBOARD
- Data Extraction from EXCEL
 - Install "xlsx" Package
 - Verify and Load the "xlsx" Package, Input as "xlsx" File
 - Reading the Excel File, Writing the Excel File
- Data Extraction from DATABASES
 - RMySQL Package, Connecting to MySql
 - Querying the Tables, Query with Filter Clause
 - Updating Rows in the Tables, Inserting Data into the Tables
 - Creating Tables in MySql, Dropping Tables in MySql
 - Using dplyr and tidyr package

Machine Learning using R

- Data Pre-processing
- Classification Algorithms
 - K Nearest Neighbors Classification
 - Naive Bayes Classification
 - Decision Tree Classification
 - Random Forest Classification
 - Support Vector Machine Classification
 - Logistic Regression
 - Kernel SVM
- Regression Algorithms
 - Simple Linear Regression
 - Multiple Linear Regression
 - Polynomial Regression
 - Support Vector Regression
 - Decision Tree Regression
 - Random Forest Regression

- Clustering Algorithms
 - K-Means Clustering
 - Hierarchical Clustering
- Association Rule Algorithms
 - Apriori
 - Eclat
- Dimensionality-Reduction
 - Principal Component Analysis
 - Linear Discriminant Analysis
 - Kernel PCA
- Model Selection & Boosting
 - Grid Search
 - K Fold Cross Validation
 - XGBoost
- Natural Language Processing
- Deep Learning - Artificial Neural Networks

DATA MINING WEKA

- Explore Weka Machine Learning Toolkit
 - Installation of WEKA
 - Features of WEKA Toolkit
 - Explore & Load data sets in Weka
- Perform Data Preprocessing Tasks
 - Apply Filters on Data Sets
- Performing Classification on Data Sets
 - J48 Classification Algorithm
 - Decision Trees Algorithm
 - K-NN Classification Algorithm
 - Naive-Bayes Classification Algorithm
 - Comparing Classification Results
- Performing Regression on Data Sets
 - Simple Linear Regression Model
 - Multi Linear Regression Model
 - Logistic Regression Model
 - Cross-Validation and Percentage Split
- Performing Clustering on Data Sets
 - Clustering Techniques in Weka
 - Simple K-means Clustering Algorithm
 - Association Rule Mining on Data Sets
 - Apriori Association Rule Algorithm
 - Discretization in the Rule Generation Process
- Graphical Visualization in Weka
 - Visualization Features in Weka
 - Visualize the data in various dimensions
 - Plot Histogram, Derive Interesting Insights